

Code-Layout, Readability and Reusability

Date	05 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-being (HDI)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the HDI – A Comprehensive Measure of Well-being project based on code structure, readability, maintainability, and reusability. A well-organized codebase with proper naming conventions, modular design, and documentation makes the project easier to understand and extend.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Properly formatted
2	Proper File Structure	Files and folders are logically organized	Yes	Organized project structure
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Easy to understand
4	Function / Method Names	Functions are descriptively named	Yes	Well-defined methods
5	Code Comments	Inline and block comments explain logic	Yes	Adequate documentation
6	Modular Design	Code is split into reusable functions/modules	Yes	Easy to maintain
7	No Redundant Code	Duplicate or unused code is removed	Yes	Optimized code
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Reliable execution

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	User authentication	Python, Flask	Login & registration	High

2	HDI Calculation	Python	HDI score calculation	High
3	Data processing module	Pandas, Numpy	Data cleaning & Analysis	High
4	Report generation	Matplotlib	Charts & reports	Medium
5	Database module	MySQL	Data storage & Retrieval	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized and modular
Readability	5	Simple and easy to understand
Reusability	5	Modules can be reused efficiently
Documentation / Comments	4	Sufficient comments provided
Overall Score	4.8	High-quality, maintainable, and reusable code